

(Please write your Enrollment Number)

Enrollment No. _____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<1>
(December, 2024)

Subject Code: BAS 101	Subject: Applied Mathematics-I (Reappear for 2021 and earlier batches)
Time : 3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) Find the eigenvalues of $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.		CO2
	b) Illustrate linear transformation.		CO2
	c) Test the convergence of $\sum \frac{1}{n+10}$.		CO1
	d) Write the Euler's formula for Fourier coefficients in a Fourier Series..		CO1
	e) Find extrema of the function $f(x,y) = x^2 + y^2$.		CO1
	f) Solve $\lim_{(x,y) \rightarrow (5,-3)} \frac{3x+2y}{x+y+1}$.		CO1
	g) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F} = 2xy\hat{i} + 3z\hat{j}$ from (0,0) to (1,1) along a curve $y = x$.		CO5
	h) Check whether $\vec{F} = (y^2 - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is solenoidal or not.		CO5
UNIT I			CO Mapping
Q2	State Cayley – Hamilton theorem. Verify Cayley – Hamilton theorem for $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$. Prove that $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$.	(10)	CO2
Q3	Find value of λ for which the following system of equations is consistent and has non-trivial solutions. Solve equations for such values of λ . $(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$, $(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$, $2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$.	(10)	CO2
UNIT II			CO Mapping
Q4	a) Determine whether the given sequence, whose n^{th} term $a_n = \frac{n^2 + (-1)^n}{n^3 + 2^n}$, is convergent or divergent. b) Discuss the convergence of the infinite series:	(5+5)	CO1

	$\frac{x}{1+x} + \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} + \dots$		
Q5	Given that the series $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \dots$ Converges to $\sin x$ for all x. a) Find the first six terms of a series for $\cos x$. For what values of x should the series converge? b) Calculate the first six terms of a series for $2 \sin x \cos x$.	(10)	CO1
UNIT III			CO Mapping
Q6	State and prove Euler's theorem for a homogeneous function $u(x, y)$. Verify Euler's theorem for $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$.	(10)	CO4
Q7	Given that $f(x, y) = x + \cos(\pi y) + x \log(y)$. Find the quadratic approximation of function $f(x, y)$ about the point $(3, 1)$.	(10)	CO3
UNIT IV			CO Mapping
Q8	Evaluate the integral by changing the order of integration $\int_0^1 \int_0^{\sqrt{2x-x^2}} dy dx$	(10)	CO5
Q9	State Gauss' s divergence theorem. Using this, evaluate $\iint_S \vec{F} \cdot d\vec{S}$, where $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and S is the surface bounded the region $x^2 + y^2 = 4$, $z = 0$ and $z = 4$.	(10)	CO5